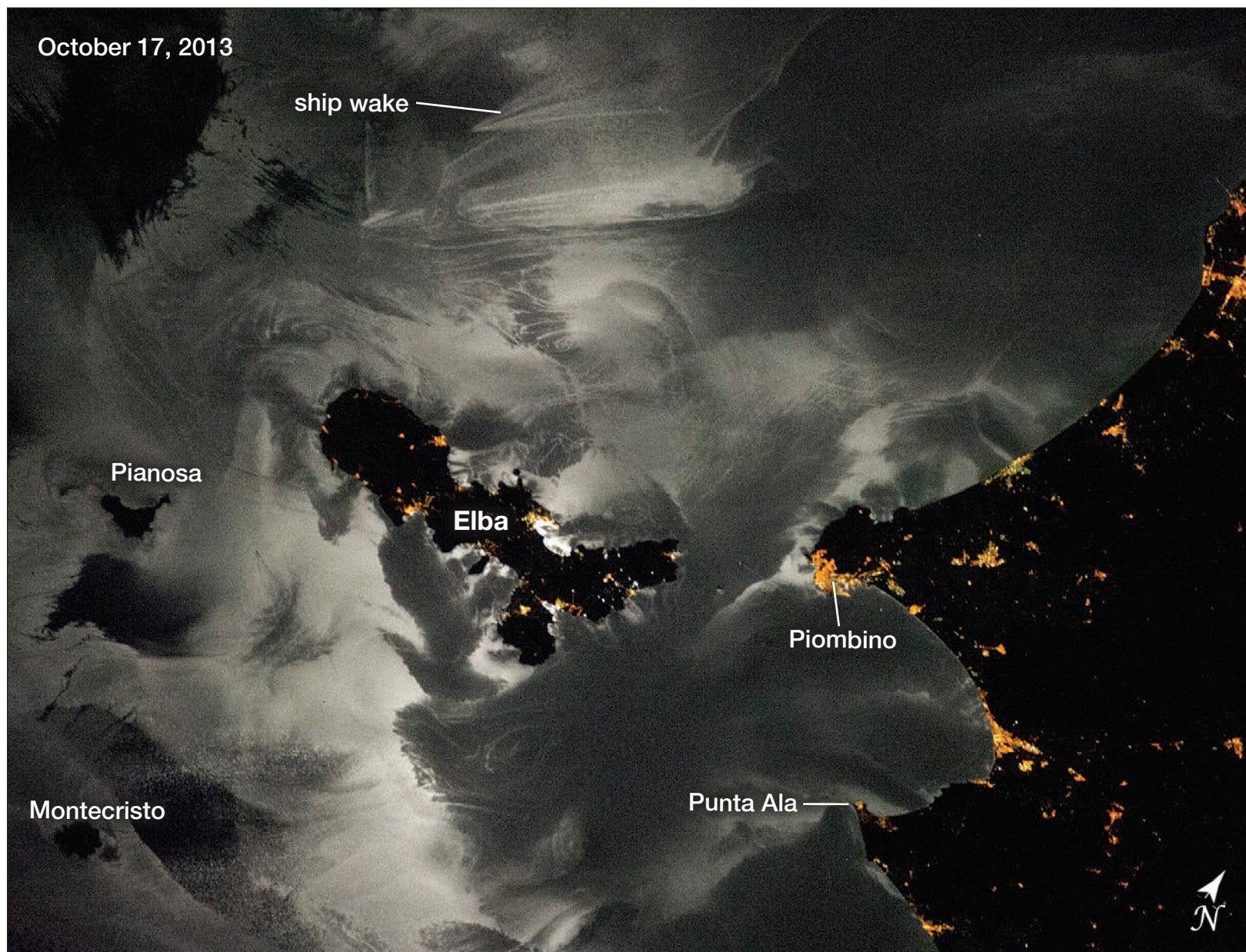


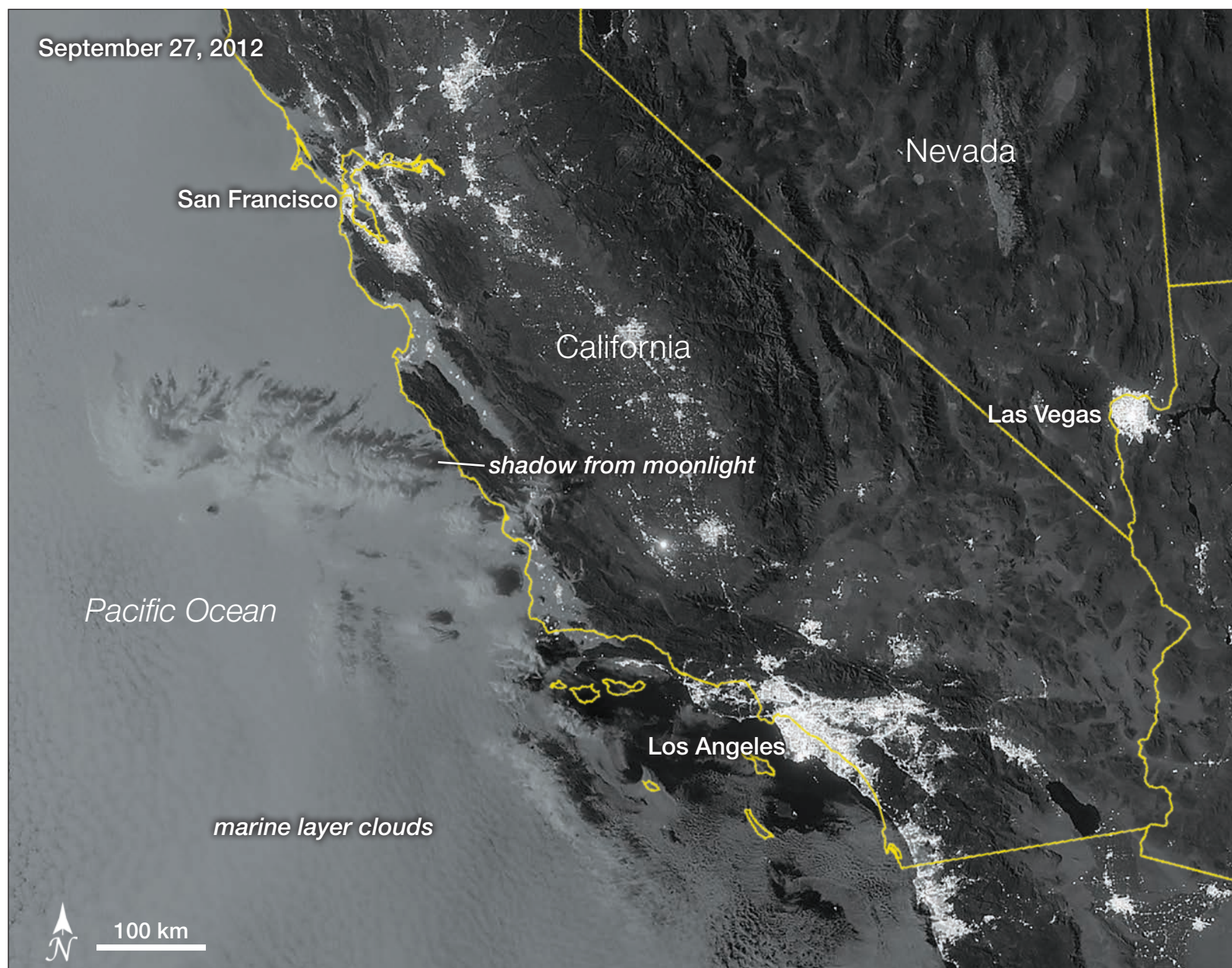




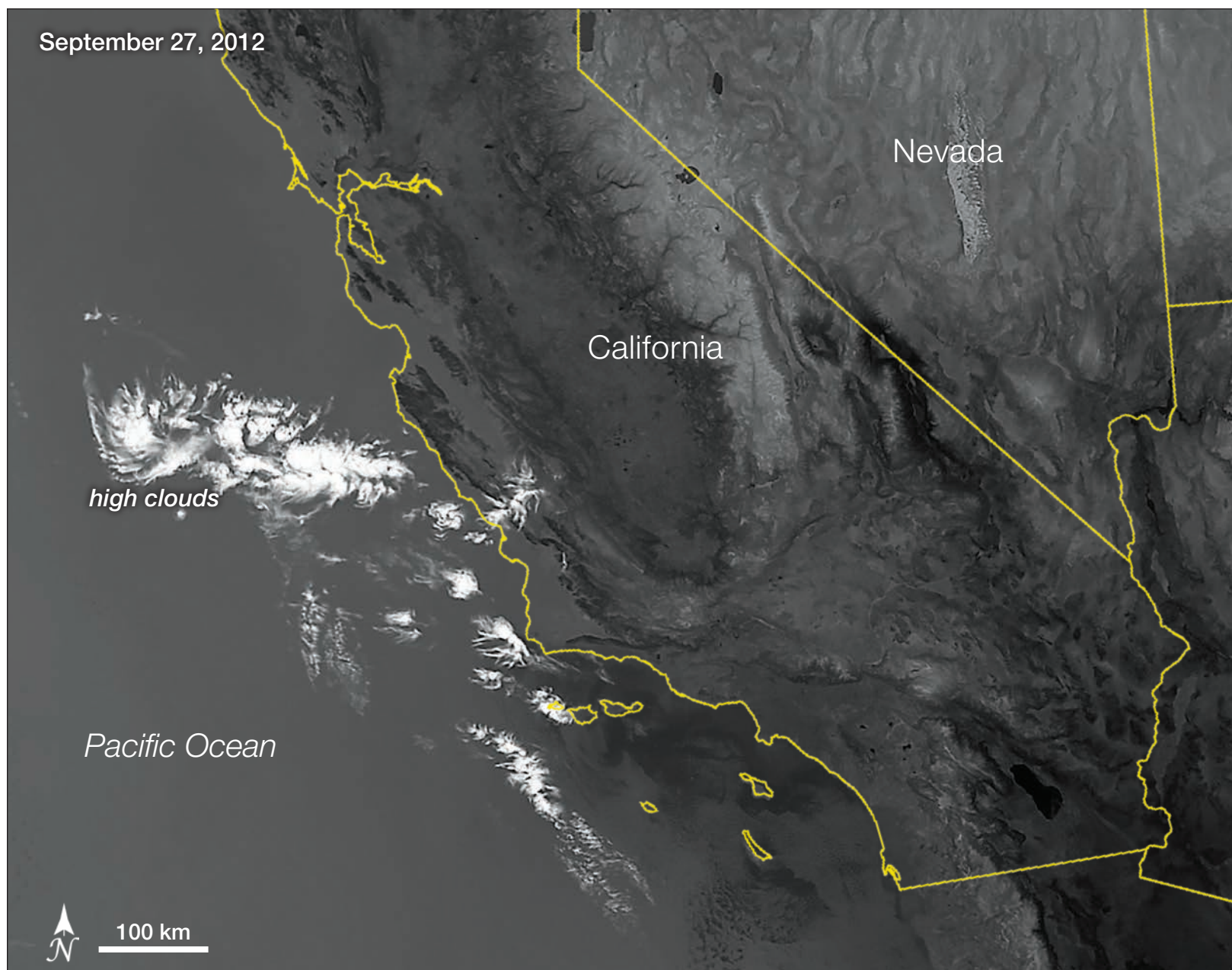
Clouds and Lightning

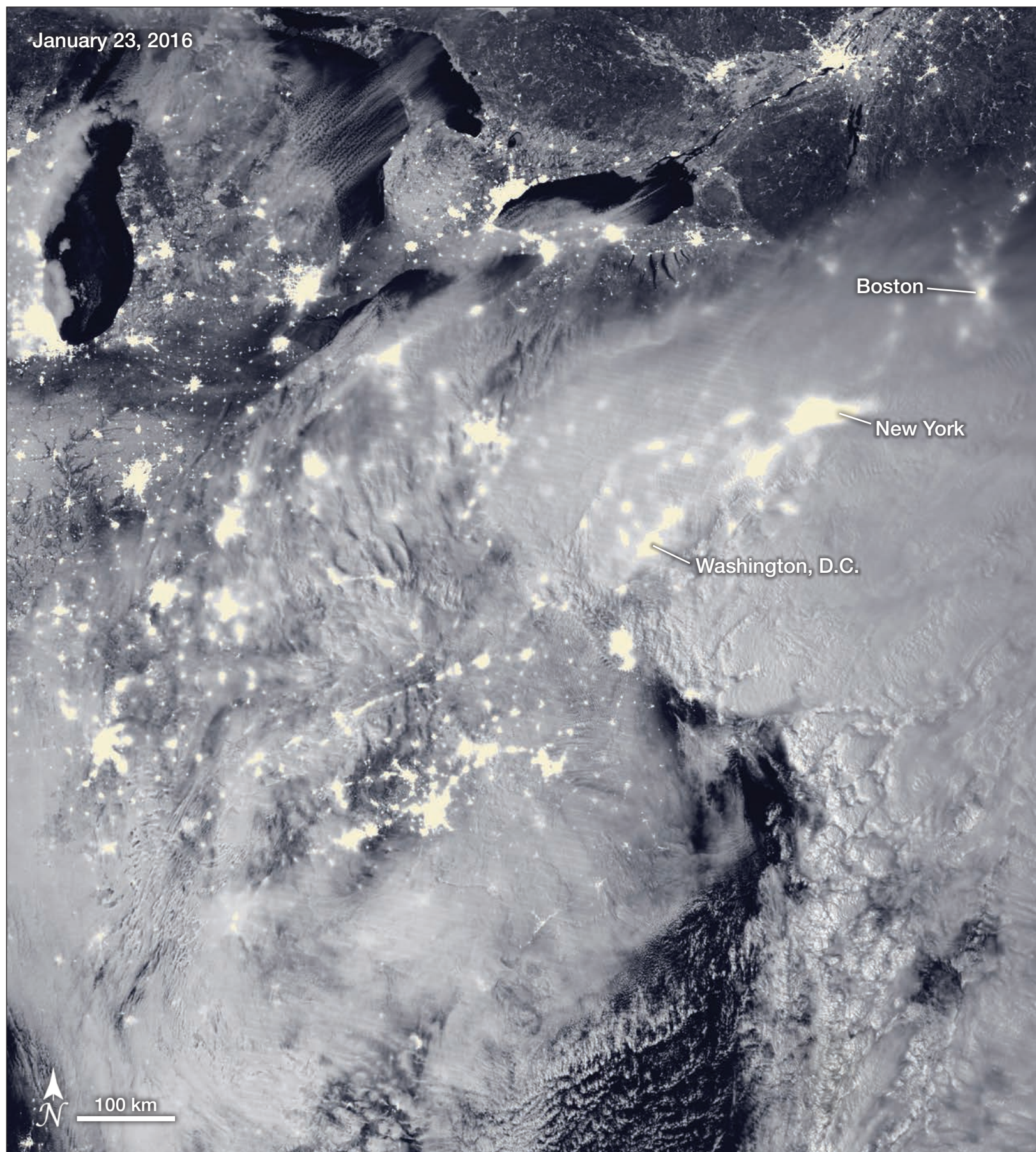
Marine Layer Clouds—California

On September 27, 2012, the VIIRS DNB on the Suomi NPP satellite captured this nighttime view of low-lying marine layer clouds along the coast of California (see below). An irregularly shaped patch of high clouds hovered off the coast of California, and moonlight caused the high clouds to cast distinct shadows on the marine layer clouds below. VIIRS acquired the image when the Moon was in its waxing gibbous phase.



Low clouds can pose serious hazards for air and ship traffic, and satellites have had difficulty detecting them in the past. To illustrate this, the image on this page shows the same scene as viewed by the VIIRS thermal infrared band used by meteorologists to monitor clouds at night. Only high clouds are visible; the low clouds do not show up at all because they are roughly the same temperature as the ground.







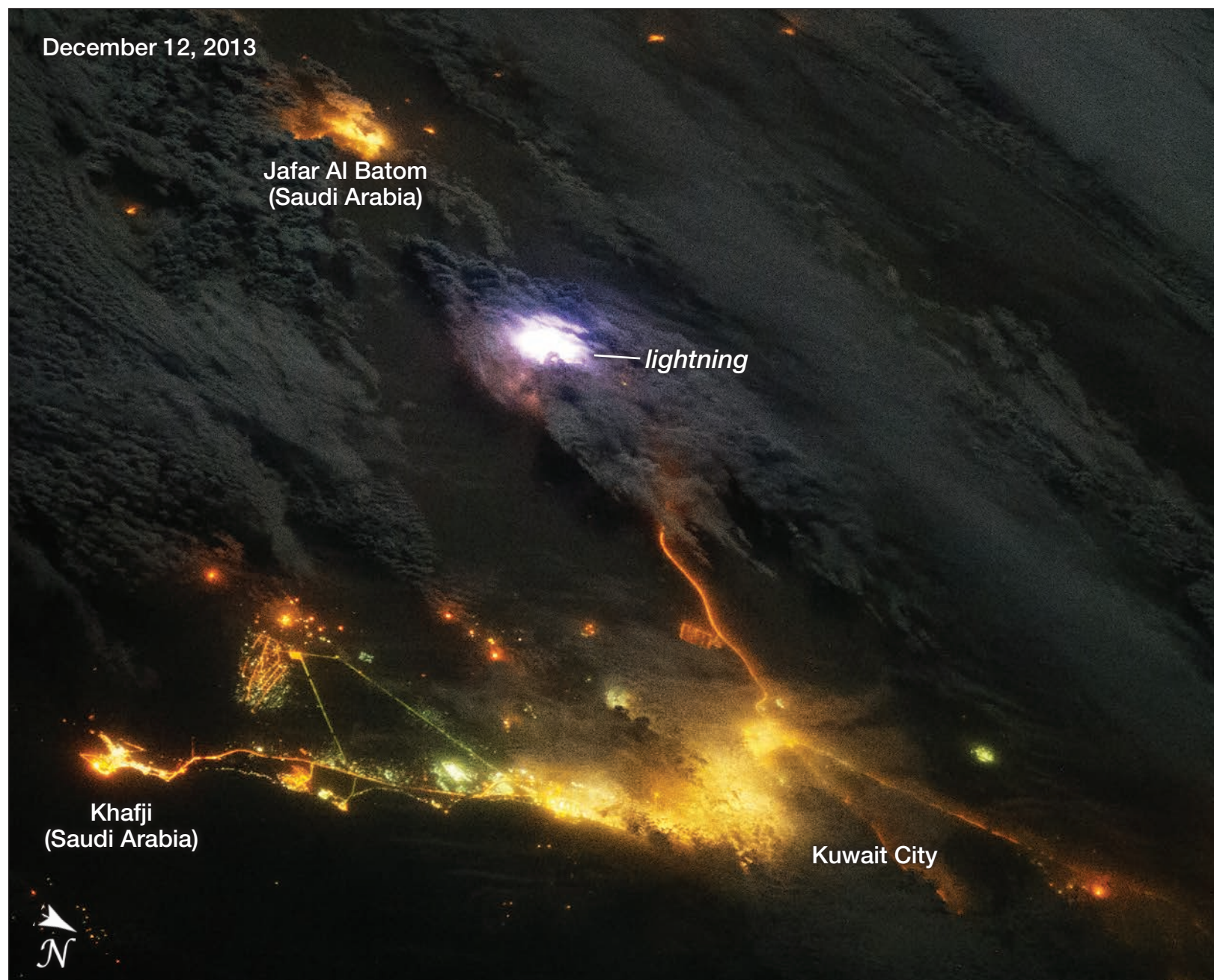
A Blizzard by Moonlight—Northeastern United States

A massive winter storm system pummeled the eastern United States in late January 2016, when two low-pressure systems merged into a potent nor'easter that dropped heavy snow from Virginia to New England. By late afternoon on January 23, snowfall totals were approaching records in several states and hurricane-force winds were battering the coastlines and leading to serious flooding. The VIIRS DNB on the Suomi NPP satellite acquired this image of the storm system at 2:15 a.m. Eastern Standard Time on January 23, 2016. Notice that the lights from the major cities along the East Coast of the United States are blurred in places by cloud cover.



Sensing Lightning from the ISS—Saudi Arabia and Bolivia

Lightning flashes about 50 times per second across all of Earth's atmosphere. That's 4.3 million times a day and roughly 1.5 billion times a year. The December 12, 2013 photograph (below, left), snapped by an astronaut onboard the ISS, shows a white flash of



lightning amidst the yellow city lights of Kuwait and Saudi Arabia. On another occasion, seen in the January 9, 2011 photograph (below, right), an astronaut onboard the ISS captured a close-up of a lightning flash beneath a thunderhead over Bolivia.



January 9, 2011

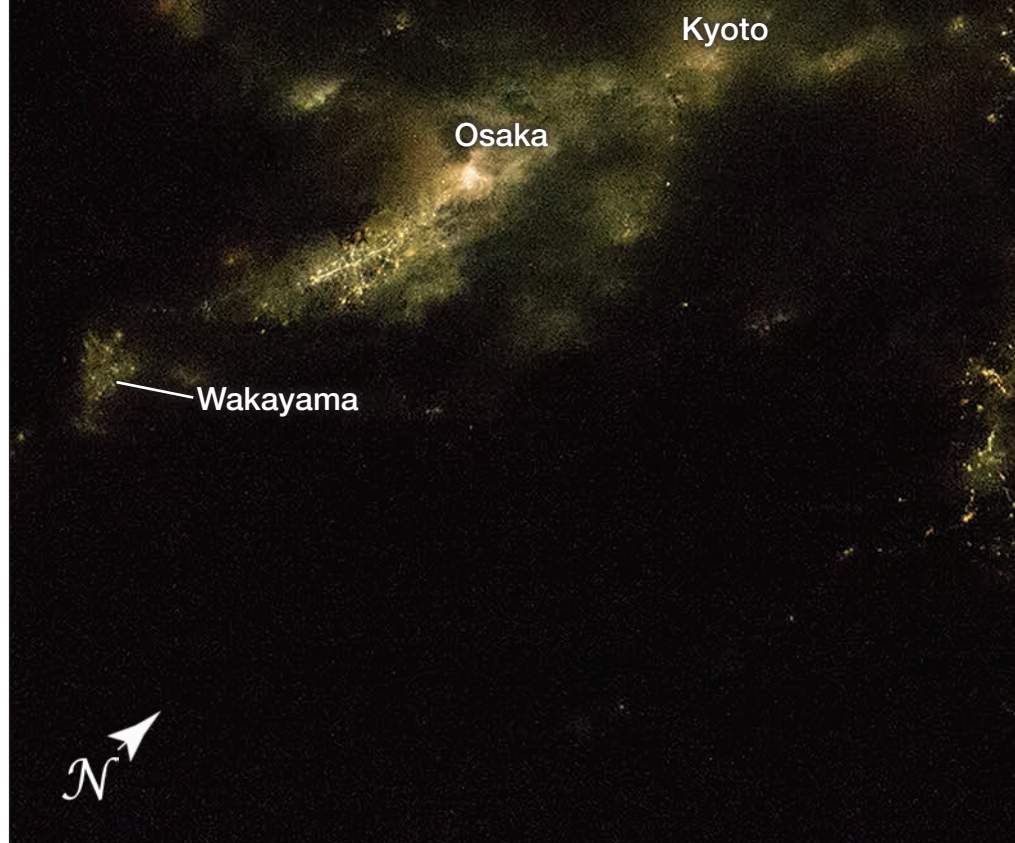




Lightning—Japan

Astronauts captured this photograph of south-central Japan on August 16, 2014, as the ISS was passing over the western Pacific Ocean. The brightest spot in the center of the photo is the port city of Nagoya, the third largest metropolitan area in Japan. To the left, the lights of the urban areas of Osaka, Kyoto, and Wakayama pierce through clearings in the cloud cover. (Note that north is to the upper right in this image.) Much of the scene is obscured by cloud cover, although a lightning storm lights up the clouds to the north. Note how the lightning is much whiter than the yellow and orange hues of city lights.

August 16, 2014





Nagoya

lightning



Elusive Sprite Captured from the ISS—Southeast Asia

They were once so elusive that scientists gave them a mystical name. Red sprites are short-lived, red flashes that occur about 50 miles (80 kilometers) up in the atmosphere. With long, vertical tendrils like a jellyfish, these electrical discharges can extend 12 to 19 miles (20 to 30 kilometers) into the atmosphere and are connected to thunderstorms and lightning.

These three serial images, part of a time-lapse movie collected from 13:41 to 13:47 UTC on April 30, 2012, show a red sprite (center image) captured with a digital camera by

